

Practice Problems

- Let $U = \text{Flowers} = \{\text{Jasmine}, \text{Rose}, \text{Lotus}, \text{Daffodil}, \text{Sunflower}, \text{Hibiscus}, \text{Chrysanthemum}\}$ be a universe on which two fuzzy sets, one of *Beautiful flowers* and the other one of *Fragrant flowers* are defined as shown below.

$$P = \text{Beautiful flowers} = \{(\text{Jasmine}, 0.3), (\text{Rose}, 0.9), (\text{Lotus}, 1.0), (\text{Daffodil}, 0.7), (\text{Sunflower}, 0.5), (\text{Hibiscus}, 0.4), (\text{Chrysanthemum}, 0.6)\}$$

$$Q = \text{Fragrant flowers} = \{(\text{Jasmine}, 1.0), (\text{Rose}, 1.0), (\text{Lotus}, 0.5), (\text{Daffodil}, 0.2), (\text{Sunflower}, 0.2), (\text{Hibiscus}, 0.1), (\text{Chrysanthemum}, 0.4)\}$$

Compute the fuzzy sets $P \cup Q$, $P \cap Q$, P^c , Q^c , $P - Q$, $P \oplus Q$. Also verify that $P \cup P^c \neq U$ and $P \cap P^c \neq \emptyset$

- Let a Mamdani model contains the following three rules:

- IF x is Small THEN y is Medium.
- IF x is Medium THEN y is Large.
- IF x is Large THEN y is Small.

where Small, Medium, and Large are fuzzy sets defined as

$$\text{Small} = 1.0/1 + 0.5/2$$

$$\text{Medium} = 0.5/2 + 1.0/3 + 0.6/4 + 0.3/5$$

$$\text{Large} = 0.4/4 + 0.7/5 + 1.0/6$$

Given the premise, x is X , where X is a fuzzy set and $X = 0.3/3 + 1.0/4 + 0.5/6$, use the max-min compositional rule of inference to compute the output of the given Mamdani model before defuzzification.